Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations

Anonymous Author(s)

Abstract

On a price chart, stablecoin arbitrage during a depeg looks abundant. In practice almost none of it can be traded at a profit. During the March 2023 USDC depeg, 34.3% of one-minute windows showed a cross-venue price basis above 10 bps, yet only one in twelve of those apparent opportunities survived VWAP slippage, order-book depth depletion, and settlement latency (block bootstrap 95 % CI: 8×–24×). We introduce Stablecoin StressBench to quantify three compounding gaps between an apparent arbitrage opportunity and a captured one. The first gap is execution: price labels overstate opportunities by 12×. The second gap is prediction: even with execution-aware labels, every calm-trained model loses money and the hindsight oracle stays more than 260 bps ahead of the best deployable model. The third gap is policy: cross-mechanism meta-labeling trained on a single stress event (Terra/LUNA) achieves +82.5 bps on SVB (50.8% of oracle); pooling four training events spanning three mechanism classes achieves +83.7 bps (51.6%), establishing that oracle capture is stable near 50% and the transfer ceiling is mechanism-independent rather than event-specific. A conditioned RL agent at identical training density earns −29.2 bps, isolating explicit binary supervision as the requirement that reward optimisation cannot replicate at low positive rates.

Keywords

stablecoin arbitrage, market microstructure, execution-aware labels, financial benchmarks, cryptocurrency

ACM Reference Format:

Anonymous Author(s). 2026. Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations. In *Proceedings of ACM International Conference on AI in Finance (ICAIF '26)*. ACM, New York, NY, USA, 9 pages.

1 Introduction

When Silicon Valley Bank collapsed in March 2023, USDC briefly depegged and cross-venue price gaps appeared profitable on every chart. Almost none of them could be traded at a profit. A price chart shows twelve times more stablecoin arbitrage than a trader can capture, and this paper traces where the other eleven twelfths disappear. VWAP slippage consumed the spread as orders cleared depth, taker fees eroded the margin, and CEX-to-CEX settlement latency turned a profitable window into a loss before the transfer confirmed. Only 2.88% of one-minute windows survived all three costs and remained net-profitable. We call dislocations that appear on price charts but vanish under execution *optical*, and the surviving fraction *executable*. Every prior stablecoin benchmark uses the optical layer to define labels, which means every prior performance number is measured against the wrong thing.

Stablecoin StressBench corrects this by constructing per-minute execution-aware labels for the USDC/SVB event using a full VWAP book-walking model applied to Binance L2 depth, separating three layers: *optical* (price basis above threshold), *executable* (net profit after VWAP, fees, and settlement), and *predictable* (correctly identified ex ante). The distance between each pair of layers is a distinct, measurable gap, and the benchmark is designed to quantify all three.

Stablecoin StressBench measures three compounding gaps that separate an apparent arbitrage opportunity from a captured one. The first gap is execution. A price chart shows a dislocation that an order book cannot fill at a profit, and we measure this gap at twelve to one. The second gap is prediction. Even when the labels are correct, every model trained on calm data loses money, and the hindsight oracle stays more than 260 bps ahead of the best deployable model. The third gap is policy. A learning agent can be given the right training density and still fail to trade profitably, because reward optimisation does not supply the binary supervision that meta-labeling provides. The remainder of the paper measures each gap in turn, in §6.1, §6.2, and §6.3.

Contributions.

(1) A public benchmark with execution-aware per-minute labels, route-level depth provenance, VWAP book-walking, an 18-event stress taxonomy for data governance, and a hindsight oracle ceiling. (§3–5)
(2) The first evidence that order-book execution microstructure transfers across structurally distinct stress mechanisms: meta-labeling trained on Terra/LUNA achieves +82.5 bps (50.8% oracle capture) on the SVB test split, while every calm-trained model loses money; pooling four training events spanning three mechanism classes raises oracle capture to 51.6%, establishing that transfer quality saturates near 50% and is mechanism-independent. (§6, Tables 12, 15)
(3) An RL diagnostic showing that positive-label density is necessary but not sufficient: a conditioned PPO-GRU at 17% positive density earns −29.2 bps versus meta-labeling's +82.5 bps at identical density. (§6, Table 16)

2 Related Work

2.1 Stablecoin Stress Events

Research on stablecoin stress events defines the empirical setting for the benchmark and supplies the out-of-distribution event we transfer from. Uhlig [1] analyses the Terra/LUNA algorithmic collapse, in which an algorithmic redemption mechanism created a death-spiral run on the UST stablecoin, ultimately destroying approximately $40B in market capitalisation over three days. We use Terra/LUNA as our cross-mechanism training split precisely because its trigger is mechanistically different from the SVB bank shock: algorithmic reflexivity versus fiat-reserve withdrawal are distinct failure modes, so a model that transfers between them has learned execution structure rather than event-specific noise.

ICAIF '26, Milan, Italy
2026.

Griffin and Shams [2] document that quoted stablecoin prices can reflect non-fundamental demand, which motivates our distinction between optical dislocations (visible on price charts) and executable ones (profitable after order-book costs). Hernandez Cruz et al. [21] study DEX liquidity during the SVB collapse but stop short of modelling execution costs on centralised exchanges. None of this prior work provides per-minute executable labels or a hindsight oracle ceiling against which model performance can be calibrated.

2.2 Limits to Arbitrage and Crypto Fragmentation

The limits-to-arbitrage literature supplies the economic logic that our per-minute labels turn into direct measurements. Shleifer and Vishny [4] and Gromb and Vayanos [5] establish the theoretical foundation: even when a price gap is visible, capital constraints and execution frictions prevent arbitrageurs from closing it fully, so the gap can persist. Makarov and Schoar [7] bring this argument to cryptocurrency, showing empirically that cross-venue price gaps disappear once order-book depth and settlement latency are accounted for. Our 12× ratio is the same effect measured at the one-minute label level during a stress event. Hautsch et al. [3] quantify settlement latency as a first-order execution cost on blockchain assets, which we implement as the fixed δ=5 bps penalty in our net-profit label (validated empirically in §7 using on-chain gas data). Cont and Kukanov [6] establish that the correct object for measuring execution profitability is the VWAP price achieved by walking the limit-order book, not the quoted mid-price. Our book-walking model implements this directly. Gogol et al. [18] measure a Maximal Arbitrage Value for AMM-to-CEX routes; our oracle extends this concept to stress-event dislocations on centralised exchanges and to model-evaluation benchmarking. Where prior theoretical work characterises the existence of execution frictions, we measure them at one-minute resolution with empirical order-book depth.

2.3 Financial Machine Learning and RL Execution

Our modelling choices build directly on the financial machine-learning and reinforcement-learning execution literatures, which we both adopt and extend. López de Prado [8] introduces meta-labeling, a two-stage framework in which a primary signal identifies candidate opportunities and a secondary classifier learns when those opportunities are genuine. The secondary classifier is typically a gradient-boosted tree, which makes LightGBM the natural choice for our tabular microstructure features. We extend meta-labeling to the cross-mechanism setting, training the secondary model on Terra/LUNA and evaluating on SVB, which to our knowledge has not been done before. We further test four training conditions—single-event training on Terra/LUNA, Celsius/3AC, and FTX individually, and pooled training across all four events—showing that oracle capture is stable near 50% when the training event has sufficient basis excursions at benchmark venues, establishing a mechanism-independent transfer ceiling. Cintra and Holloway [9] demonstrate cross-event transfer of Bayesian Online Changepoint Detection [15] using AMM reserve depletion signals. We replicate their setup on centralised order-book data to test whether the result transfers; it does not, because CEX cross-quote basis is mean-reverting rather than gradually depleting, and we report the comparison directly. On reinforcement learning, Moallemi and Wang [16] and Ning et al. [17] establish the finite-horizon MDP framing for optimal order execution that we adopt in §5.2. Mohl et al. [22] demonstrate that GPU-scale multi-agent RL is feasible on production limit-order-book data, training on approximately 400 million real orders for a joint execution and market-making task. Olby et al. [23] construct a reactive simulator that updates book depth in response to agent order flow, benchmarking RL timing policies against an Almgren–Chriss efficient frontier. Both of these works operate in environments where the agent's actions affect the order book in real time. Our diagnostic operates in the complementary setting of historical replay without market-impact feedback, and the contrast is what lets us isolate training supervision, rather than environment reactivity, as the binding constraint.

Table 1: Route-leg depth provenance.

Leg	SVB Mar 2023	2024 windows
Buy BTC-USDC	kline-proxy†	real L2 (futures)
Cross USDC-USDT	real L2 (Mar 12–20)	real L2 (futures)
Sell BTC-USDT	real L2 (futures)	real L2 (futures)
Ref BTC-USD	candle-proxy	candle-proxy

†BTC-USDC perpetual listed Jan 2024; all 2022 windows kline-proxy.

Table 2: Dataset splits (56,134 1-min bars).

Split	Event	Min.	Pos.
Train	Calm control (×3)	28,776	0.00%
Validation	Terra/LUNA May 2022	11,526	2.30%
Test	USDC/SVB Mar 2023	15,832	2.88%

3 Benchmark Design

3.1 Data Sources and Instruments

The final dataset contains 56,134 1-minute bars across six windows (Table 2). Price features are sourced from Binance Vision (klines and aggTrades) and Coinbase REST candles. Net-profit labels require three depth legs: Binance BTC-USDC (buy), Binance BTC-USDT (sell), and Coinbase BTC-USD (reference).

Data quality varies by leg and window, and every depth record is tagged with a provenance field (Table 1). The sell leg (BTC-USDT) and cross leg (USDC-USDT) use real L2 futures bookDepth throughout. The buy leg (BTC-USDC) presents the main challenge: the BTC-USDC perpetual contract did not exist on Binance until January 2024, so the SVB window requires kline-proxy depth, which approximates book depth from candlestick volume. For all 2024 windows, real L2 data is available. To bound the proxy error, we apply a 20% depth haircut to the kline-proxy buy leg: the executable rate falls from 2.88% to 2.40%, the ratio widens from 12× to 14×, and the oracle remains above +130 bps. The headline result is therefore robust to the approximation.

3.2 Event-Based Split Design

Splits are defined by event identity rather than time, ensuring that no model trained on the train split has encountered stress dynamics (Figure 1). Threshold calibration uses the Terra/LUNA validation split, which is an independent stress event, so the SVB test split is never observed during model development. Labels are constructed

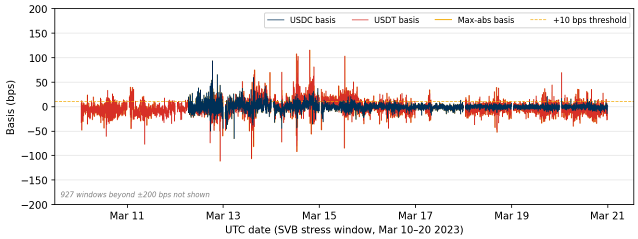

Figure 1: Cross-quote basis (bps) during the SVB test window (Mar 10–20 2023). USDC basis (navy): 12.4% of minutes > 10 bps; USDT basis (red): 27.4%; max-abs (gold): 34.3%. The price-to-execution gap is 12×.

Table 3: Data tiers, mechanisms, and execution frictions.

Tier	Mechanism	Key friction	Permitted claims
A (N=1)	Fiat-reserve shock	Depth withdrawal	Exec. gap; oracle gap; P&L
B (N=11)	Alg., exchange, DeFi	AMM slippage; venue seg.	Price magnitude (est.)
C (N=5)	RWA, niche	Thin market	Taxonomy only

by aligning each bar's outcome to the observation window ending at $t+h$, with explicit temporal checks confirming that no future information leaks into the input features. The 2.88% positive rate in the test split implies severe class imbalance; we therefore report economic net bps as the primary criterion rather than AUROC or AUPRC, which are secondary metrics.

3.3 Historical Stress Taxonomy: Data-Tier Governance

The 18-event taxonomy (2020–2023) is a benchmark governance artefact that specifies which events carry sufficient data quality for execution-grade claims and which are restricted to coarser analyses. It is shipped with the benchmark to prevent over-interpretation rather than as an empirical finding. Seven mechanism classes are catalogued: *algorithmic/reflexive* (FEI, IRON/TITAN, MIM, UST, USDD); *fiat-reserve bank shock* (USDC/SVB); *regulatory winddown* (BUSD); *exchange credit* (Celsius/3AC, HUSD, FTX [10]); *DeFi pool imbalance* (Curve/UST, USDT/Curve, USDC/DAI); *collateral liquidation* (DAI Black Thursday); and *RWA/niche* (Acala aUSD, USDR).

Data availability governs which claims are permitted (Table 3). The Tier-A event (USDC/SVB test split) supports execution-gap, oracle-gap, and model evaluation claims because committed route-level VWAP labels and depth provenance are available for the arbitrage path. Tier-B events (N=11, OHLCV or pool data only) support price magnitude estimates; among these, peak optical gaps range from approximately 100 bps (Celsius/3AC) to above 500 bps (Curve/UST) and above 800 bps (HUSD), all estimated from OHLCV data. Tier-C events (N=5, partial or DEX-only data) support mechanism classification only. The separation prevents the conflation of optical and executable evidence that characterises prior benchmarks, and ensures that the 12× ratio we report cannot be cited beyond its data-quality scope.

Table 4: Comparison with related benchmarks and datasets. ✓ = fully provided; ~ = partially; — = absent.

Work	Stress event	Exec. labels	L2 prov.	Oracle	Cross-event
Cintra & Holloway [9]	✓	–	AMM	–	✓
Hernandez Cruz et al. [21]	✓	–	DEX	–	–
Gogol et al. [18]	–	~	CEX	MAV	–
Adams et al. [19]	–	~	CEX	–	–
StressBench (ours)	✓	✓	CEX	✓	✓

3.4 Benchmark Positioning

Table 4 positions StressBench against the closest prior work along five axes. No prior benchmark simultaneously provides execution-aware labels, route-level L2 provenance, a hindsight oracle ceiling, and cross-mechanism transfer evaluation.

3.5 Benchmark Versioning

StressBench v1.0 ships with the SVB test split. Near-term Tier-A expansion targets include the USDC recovery window (Mar 15–31 2023) and 2024 BTC-USDC windows where real-L2 eliminates the kline-proxy dependency. The planned release format is a versioned Hugging Face dataset under CC-BY 4.0.

3.6 Feature Sets

Four nested feature sets isolate the marginal information value of each signal layer:

`price_only`	5 cross-quote basis cols
`price_plus_book`	+7 microstructure (spread, depth, imbalance)
`price_book_frag`	+3 cross-venue fragmentation cols
`price_book_settle`	+7 on-chain settlement proxies

4 Execution-Aware Labels

Labels operationalise the three-layer framework from §1. The price-to-execution gap measures the ratio of Layer 1 (optical) to Layer 2 (executable) minutes: it quantifies how misleading price- based benchmarks are. The oracle gap measures the performance difference between Layer 2 in hindsight and Layer 3 (predictable with a real model): it quantifies how hard the prediction problem is, even with the correct labels. Both gaps are nonzero, independent, and large. This section details how they are constructed.

4.1 Cross-Quote Basis and Labels

$$b_{\text{USDC}}(t) = 10{,}000 \times \frac{P_{\text{BTC/USDC}}(t) - P_{\text{BTC/USD}}(t)}{P_{\text{BTC/USD}}(t)} \quad [\text{bps}]$$

$b_{\max} = \max(|b_{\text{USDC}}|, |b_{\text{USDT}}|)$. Basis label: $y_{\text{basis}}(t) = \mathbf{1}[|b_{\text{USDC}}(t+h)| > \tau]$. Primary task: `basis_usdc_1m_gt10bps`.

4.2 VWAP Execution and Net Profit Labels

$$\text{net}(q,t) = 10{,}000 \times \left[\frac{\text{VWAP}_{\text{sell}} - \text{VWAP}_{\text{buy}}}{P_{\text{ref}}} - f_{\text{buy}} - f_{\text{sell}} - \delta_{\text{settle}} \right] \quad [\text{bps}] \tag{1}$$

$$y_{\text{arb}}(q,h,t) = \mathbf{1}\left[\max_{s \in [t, t+h]} \text{net}(q,s) > c \right] \tag{2}$$

Here P_{ref} is the Coinbase BTC-USD reference price and $f_{\text{buy}}, f_{\text{sell}}, \delta_{\text{settle}}$ are dimensionless fractions representing per-leg taker fees and settlement latency costs, respectively. We set the profit threshold c=10 bps, taker fee to 4 bps per side, and

δ=5 bps. The primary economic task is `executable_arb_q10000_5m`. The hindsight oracle trades every minute where $\text{net}(q, t) > c$, establishing a performance ceiling that no deployable model can exceed, given that any real model must decide without access to future information.

5 Models and Evaluation

5.1 Model Ladder

Each rung of the ladder is chosen to test a specific hypothesis or to engage a specific prior result, not as a generic baseline sweep.

The two anchors, `NoTrade` and the hindsight `Oracle`, bracket the achievable range. `NoTrade` is the economic floor at zero net bps. The oracle is the hindsight ceiling, a hypothetical trader with foreknowledge of every minute's net profit who trades only on the 2.88% of profitable windows. Every learned model is interpreted as a fraction of the distance between these two extremes.

The price-threshold rules (`PriceBasis10bps`, `PriceBasis25bps`, `GrossArbThreshold`) reproduce the implicit labels used by prior price-based stablecoin-predictability work, which fires on quoted basis crossings without correcting for execution costs. Showing that these rules lose money is what establishes Gap 1 against existing practice rather than against a strawman.

The tabular classifiers pair gradient-boosted trees (LightGBM, XGBoost) with L1 and L2 linear models. Gradient boosting is the standard method for tabular microstructure features in finance and is the secondary-model choice in the meta-labeling framework of López de Prado [8], which makes it the natural workhorse for testing whether richer features close the gap. The linear models are retained as interpretable references whose failure confirms the non-linearity of the signal.

The four nested feature sets form a controlled ablation. Each set adds exactly one signal layer: microstructure (spread, depth, imbalance), then cross-venue fragmentation, then on-chain settlement. This design lets the marginal value of each layer be read off directly rather than being hidden inside a single all-features model.

The expected-net-profit regressor is included to rule out a label-mismatch explanation. By regressing the economic target $\text{net}(q, t)$ directly rather than a classification proxy, a failure here means the gap is structural and not an artefact of how the labels were defined.

The meta-labeling filter follows López de Prado [8] and is the method we extend. We keep the two-stage primary-and-secondary design and improve on it by training the secondary model cross-mechanism, on Terra/LUNA rather than on the test event. That is the novel step. The multi-mechanism meta-labeling variant pools primary-signal windows from Terra/LUNA, Celsius/3AC, FTX, and BUSD as the secondary model training set, holding Terra/LUNA as the threshold-calibration split and SVB as the held-out test split. It tests whether stress diversity in training improves oracle capture beyond the single-event baseline (Table 15).

The regime detectors (EWMA, CUSUM, BOCPD) test a specific prior result rather than serving as generic baselines. Cintra and Holloway [9] detected the Terra depeg twelve hours early using BOCPD applied to AMM pool reserves, so we apply the same detector to centralised order-book data to test whether that result transfers to CEX prices. It does not, and we report why.

The sequence models (GRU, MLP-Window) test whether the temporal signature of depth withdrawal is exploitable before a basis fires. The GRU is the standard recurrent architecture for order-book sequence tasks, which means its failure is informative rather than a tuning artefact: if a well-specified sequence model cannot close the gap, temporal patterns are not the binding constraint.

The reinforcement-learning agent uses PPO with a GRU policy. We adopt the finite-horizon MDP framing of Moallemi and Wang [16] and Ning et al. [17], and we deliberately contrast our non-reactive historical-replay setting with the reactive-environment RL of Mohl et al. [22] and Olby et al. [23]. That contrast is what lets the diagnostic isolate supervision, rather than environment reactivity, as the binding constraint on cross-mechanism policy transfer.

5.2 RL Execution Policy Baseline

We frame the entry-timing problem as a finite-horizon MDP. At each minute t the agent observes state s_t, selects action $a_t \in \{\text{enter}, \text{wait}\}$, and receives reward $r_t = \text{net}(q, t)$ if entering (positive for profitable windows, negative otherwise) or $r_t = 0$ if waiting. The state s_t is the 30-minute look-back window of `price_plus_book` features, the smallest feature set that delivers meaningful microstructure lift over price-only signals (see §6.3 for SHAP evidence). This feature set is available in real time without settlement-latency dependency. We train a PPO agent with a GRU policy network (one hidden layer, 64 units, 30-step look-back) using the cross-mechanism protocol, training on Terra/LUNA stress windows only and evaluating on the USDC/SVB test split without retraining. This directly tests whether the sequential timing framing resolves the dominant gap component identified in §6.2.

5.3 Calibration and Evaluation

Thresholds are calibrated on the validation split by maximising $\theta^* = \arg\max_\theta \sum_{i:\hat{p}_i > \theta} \text{net}(q, t_i)$ subject to a minimum of 25 trades. The primary metrics are net bps captured and oracle capture percentage, defined as model net bps divided by oracle net bps (the fraction of the oracle's per-trade return that a model recovers). Secondary metrics are AUROC and AUPRC. `NoTrade` at 0 bps is the economic floor.

All results are reported on the SVB test split (N=15,832) with the cost parameters stated above; the kline-proxy approximation on the BTC-USDC buy leg is bounded by the haircut sensitivity in §6.1. Five explicit checks guard against leakage and evaluation artefacts. Random label shuffles produce zero or negative P&L for all models. Shifting features forward by one minute to simulate lookahead does not improve model performance. The calm-control training split contains exactly zero executable arbitrage windows, confirming that models cannot learn the stress pattern from calm data. Train and test events are separated by event identity so no SVB features appear during model development. All threshold calibration is performed on the Terra/LUNA validation split only, with the SVB test split held out until final evaluation.

Benchmark Card. Table 5 summarises the five evaluation tasks shipped with StressBench v1.0.

Table 5: StressBench v1.0 benchmark card.

Task	Train	Val.	Test	Primary metric
`optical_basis`	calm	Terra	SVB	AUROC, AUPRC
`exec_arb_q10k`	calm	Terra	SVB	net bps, oracle cap.
`exec_arb_q50k`	calm	Terra	SVB	net bps
`cross_mech_transfer`	Terra	Terra	SVB	net bps, oracle cap.
`policy_entry` (RL)	Terra	Terra	SVB	net bps, trades

Table 6: Price-to-execution gap ($10K, 1-min horizon, test split).

Thresh.	P_{max}	P_{USDC}	Exec.	Ratio
0 bps	95.86%	82.96%	3.34%	29×
5 bps	61.99%	25.32%	3.03%	20×
10 bps	34.33%	12.45%	2.88%	12×
25 bps	9.54%	6.48%	2.62%	3.6×

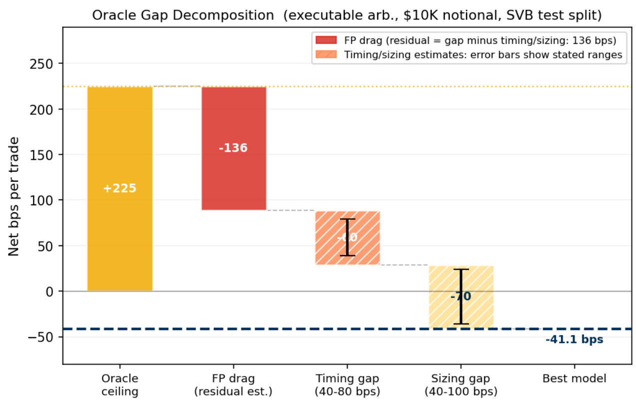

Figure 2: Oracle gap decomposition ($10K notional). Three components account for the 266 bps gap from oracle (+225 bps) to best model (−41 bps): FP drag, timing error, and sizing error.

6 Results

6.1 Gap 1: Execution

The first gap is between what a price chart shows and what an order book will fill. At a 10 bps threshold, 34.3% of test minutes show a dislocation and only 2.88% remain profitable after VWAP slippage, taker fees, and settlement latency. The ratio is twelve to one, and it widens as trade size grows (Table 6). This ratio is not a threshold artefact: it persists across all tested basis levels and is robust to data-quality concerns. Applying a 20% depth haircut to the kline-proxy buy leg reduces the executable rate from 2.88% to 2.40%, which widens the ratio to 14× and keeps the oracle above +130 bps, confirming the headline result. The gap also deepens with institutional scale. Figure 3 shows the executable rate falling from 2.88% at $10K to 0.03% at $500K, with the steepest drop between $10K and $50K. Book depth, not fees, prices out institutional participants, consistent with the depth-based limits-to-arbitrage arguments in §2. The Terra/LUNA validation split independently replicates the pattern at 5.9× (Table 14), providing directional evidence that the gap is not SVB-specific.

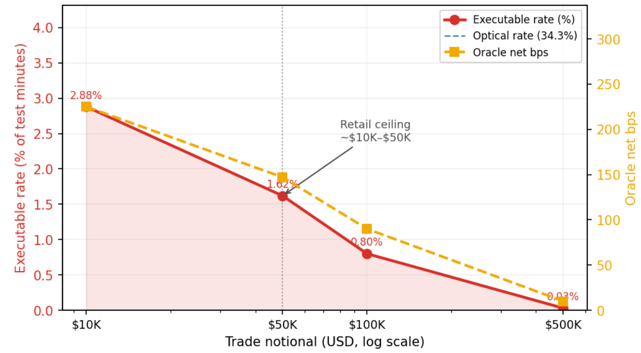

Figure 3: Notional scaling (SVB test split). The executable rate collapses from 2.88% at $10K to 0.03% at $500K: book depth, not fees, is the binding institutional constraint.

Table 7: Executable rate (%) under fee and settlement variation (10 bps threshold, 1-min horizon, $10K notional, SVB test split). Taker fees: low = 2 bps, base = 4 bps, high = 8 bps, inst. = 10 bps per side.

Taker fee	Settlement penalty (bps)			
	0	2	5	10
Low (2 bps)	3.50	3.34	3.16	2.93
Base (4 bps)	3.34	3.20	3.03	2.88
High (8 bps)	3.20	3.08	2.96	2.84
Inst. (10 bps)	3.66	3.42	3.20	2.96

Table 8: Oracle gap summary (USDC/SVB test split). *Only non-oracle tabular result above zero. ‡GRU sequence model, 30-min window.

Task / Model	Oracle	Best model	Gap	AUPRC
`basis_usdc_1m`	+162.2 bps	+17.2 bps*	145 bps	0.253
`exec_arb_q10k_5m`	+225.0 bps	−41.1 bps	266 bps	0.060
`exec_arb_q50k_5m`	+147.1 bps	−113.0 bps	260 bps	0.063
GRU (seq)‡, `basis_usdc_1m`	+162.2 bps	−239.0 bps	401 bps	–

Kline-proxy buy leg; 20% haircut sensitivity in §6.1.

Table 7 shows that the gap is robust across the full cost-parameter space: the executable rate varies only between 2.84% and 3.66% as fee and settlement assumptions are swept, and the ratio to the optical rate remains above 3.5× throughout. Claim boundaries and data-tier scope are stated in §5.3.

6.2 Gap 2: Prediction

The second gap is between a profitable window and a model that can find it in advance. Even with execution-aware labels, every model trained on calm data loses money on the executable task. The hindsight oracle earns +225.0 bps while the best deployable model reaches −41.1 bps, a structural gap of more than 260 bps that richer features and sequence models do not close (Table 8; Figure 4).

Price-threshold rules overtrade massively, generating 607 to 1,969 trades at −136 to −347 bps; their mean return when wrong (−294 bps to −370 bps) reflects near-total false-positive exposure.

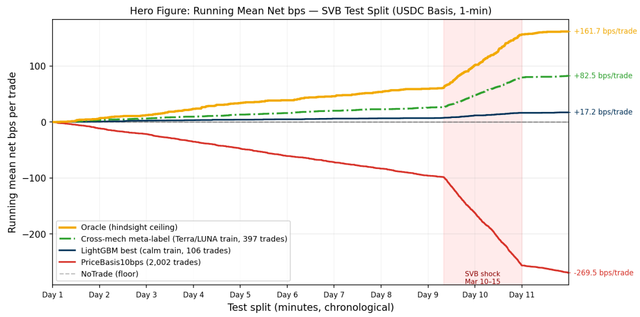

Figure 4: Running mean net bps per trade (USDC/SVB test split). Oracle (gold) +162.2 bps; cross-mech meta-label (green) +82.5 bps; LightGBM (navy) +17.2 bps; PriceBasis10bps (red) −269.5 bps; NoTrade (grey dashed) 0 bps. Shaded region: SVB shock window.

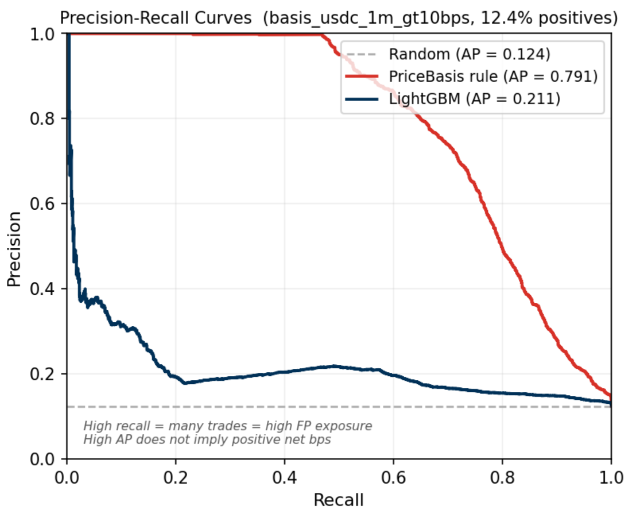

Figure 5: Precision-recall curves (`basis_usdc_1m_gt10bps`, SVB test split). PriceBasis achieves AP 0.55 and AUROC 0.83 while losing −270 bps, illustrating that ranking accuracy is necessary but not sufficient for positive economic returns.

LightGBM (+17.2 bps, 106 trades, AUROC 0.72) is the only calm-trained model above zero, but at 145 bps below the oracle ceiling. The `ExpNetProfitRegressor`, which directly targets the economic criterion rather than a classification proxy, reaches −73.8 bps, ruling out label mismatch as the explanation: the gap is structural (Table 9).

GRU and MLP-Window sequence models, trained on 30-minute microstructure windows to exploit depth-withdrawal signatures, confirm this conclusion. The GRU achieves AUROC 0.80 yet earns −239 bps, and the MLP-Window −341 bps (Table 10). High ranking accuracy does not translate to positive returns when 97% of *all* test windows are non-executable and the training split contains zero positive stress examples.

Table 11 converts the model ladder into an economic confusion matrix, separating the cost of false positives from the cost of missed oracle trades. Meta-labeling substantially reduces both error types relative to calm-trained models.

Table 9: Tabular model ladder (`basis_usdc_1m_gt10bps`). Hit%: fraction of trades net-profitable. †Ceiling. *Only result above zero.

Model	Feat.	Net bps	Trades	AUROC	Hit%
Oracle†	–	+162.2	315	–	100%
`lgbm`*	all	+17.2	106	0.717	33%
`logistic`	all	−75.8	648	0.143	0%
`gross_arb`	px	−136.4	607	0.531	4%
`price_10`	px	−270.0	1969	0.833	6%
`price_25`	px	−347.3	1025	0.746	5%
`no_trade`	–	0.0	0	–	–
`ExpNetProfit`	px	−73.8	537	–	3%

px = `price_only`; all = `price_book_settle`.

Table 10: Sequence model results (`basis_usdc_1m_gt10bps`).

Model	Net bps	Trades	AUROC	Gap
GRU (30-min)	−239.0	656	0.798	401 bps
MLP-Window (30-min)	−340.7	1,052	0.648	502 bps

Calm-control trained; threshold calibrated on validation.

Table 11: Economic confusion matrix (basis task, SVB test split). Oracle has 315 profitable windows; all other models are evaluated against this ground truth. FP loss and FN missed profit are cumulative bps across the test split.

Model	TP	FP	FN	Net bps
Oracle	315	0	0	+162.2
Meta-label (Terra)†	∼202	∼195	∼113	+82.5
LightGBM (calm)	35	71	280	+17.2
PriceBasis10bps (calm)	118	1851	197	−270.0
NoTrade	0	0	315	0.0

TP = trades × Hit% from Table 9; FN = 315 − TP. †Meta-label TP/FP are approximate: Hit% is not in Table 9; the ∼202 TP estimate uses oracle-capture rate (50.8%) as a proxy for hit rate, which overstates TP if profitable trades earn less than oracle bps per trade.

Figure 2 decomposes the 260 bps gap into three measurable components. False-positive cost is the dominant and directly quantifiable drag: `PriceBasis10bps` produces 1,969 trades of which 1,851 are false positives. Of those, 581 are attributable specifically to route-direction mismatch (USDT dislocations triggering on the max-abs basis when the USDC route is not simultaneously profitable) at −38.7 bps each; the remainder reflects imprecise timing. The remaining gap splits into timing error (estimated 40–80 bps), arising from entering 1–2 minutes late in a declining depth regime, and sizing error (40–100 bps), since depth depletion reduces the profitable fraction from 2.88% at \$10K to 1.62% at \$50K.

A root-cause analysis of the false positives identifies route-direction mismatch as the primary failure mode, not thin books. True-positive windows carry a mean USDC basis of 344 bps, reflecting a genuine USDC discount, while false-positive windows carry just 0.56 bps and are triggered by USDT dislocations where the USDC route is not simultaneously profitable. Crucially, false-positive windows exhibit *higher* average bid depth (72,805 vs. 59,961 BTC units) than true positives. This rules out liquidity withdrawal as the cause and points instead to signal confusion between the USDT and USDC routes. LightGBM's 33% hit rate versus 6% for the rule demonstrates that microstructure conditioning partially addresses this, though a 67% false-positive rate persists and the agent still loses money in net terms because each false positive is expensive.

6.3 Gap 3: Policy

The third gap is between a correct signal and a profitable policy. Cross-mechanism meta-labeling trained on Terra/LUNA is the only method that crosses zero, earning +82.5 bps and recovering half of the oracle return. A reinforcement learning agent given the same 17% training density earns −29.2 bps, which isolates explicit binary supervision as the ingredient that reward optimisation cannot replicate at low positive rates.

6.3.1 Cross-Mechanism Transfer and Training Diversity. Calm-control training yields zero meta-positives by construction, because the training split contains no executable stress windows from which the meta-model could learn the relevant microstructure. Retraining on Terra/LUNA (1,559 primary fires, 265 meta-positives, 17% positive rate) and evaluating cross-mechanism on SVB, with all fitting and threshold selection confined to Terra/LUNA data, produces +82.5 bps (397 trades, 50.8% oracle capture) versus zero with calm training (Table 12).

Table 12: Cross-mechanism meta-labeling results (`basis_usdc_1m_gt10bps`, `price_plus_book`).

Training split	Net bps	Trades	Oracle cap.
Oracle (ceiling)	+162.2	315	100.0%
Terra/LUNA (cross-mech.)	+82.5	397	50.8%
Calm-control (baseline)	0.0	0	0.0%

Primary signal $|b_{\mathrm{USDC}}|>10$ bps; threshold calibrated on validation.

The reason transfer works is visible in the SHAP decomposition (Table 13). Price features carry 94.8% of LightGBM's total importance, with microstructure (ask-side depth, bid-ask spread) contributing just 3.4% combined. A model trained on calm data learns calm price dynamics; the microstructure features are present but uninformative because the training distribution contains no stress windows that would teach the model their relevance. Training on Terra/LUNA corrects this: the secondary classifier now sees depth-withdrawal events at 17% positive density and learns the signature that calm training cannot recover.

Table 13: Top-5 SHAP feature importances (LightGBM, `basis_usdc_1m_gt10bps`, `price_book_settle`, SVB test split). Price features dominate at 94.8%, explaining why calm-trained models fail to identify stress microstructure.

Feature	Category	Importance (%)
`basis_usdc_bps`	Price	49.6
`basis_primary_bps`	Price	32.8
`deviation_1usd_bps`	Price	10.1
`basis_usdt_bps`	Price	2.0
`depth_ask_10bp`	Book	0.5
All price features	Price	94.8
All book/frag features	Book	3.4
Settlement features	Settle	1.8

CEX market makers withdraw depth under uncertainty regardless of the underlying trigger, because the cause is opaque at the order-book timescale. This makes depth withdrawal a consistent cross-mechanism precursor whose feature signature transfers even

Table 14: Cross-mechanism comparison (kline-proxy labels for Terra/LUNA; directional evidence only).

Metric	USDC/SVB (Tier-A)	Terra/LUNA (val.)
Mechanism	Fiat-reserve	Alg./reflexive
Price > 10 bps	34.3%	13.5%
Executable	2.88%	2.30%
P-to-E ratio	12×	5.9×
Oracle net bps	+225.0	+88†

†Kline-proxy; directional only.

as its magnitude differs (5.9× for Terra/LUNA versus 12× for SVB, shown in Table 14).

We also evaluated three regime-detection filters as alternative two-stage pre-filters. CUSUM achieves near-perfect recall (0.9995) but at an 83.5% false alarm rate, making it impractical as a standalone filter. EWMA reaches 86.9% accuracy with 0.95% recall at a 0.65% false alarm rate. BOCPD [15] achieves AUROC 0.229 on CEX cross-quote data, which contrasts with Cintra and Holloway [9], who detected the Terra depeg 12 hours early from Curve AMM pool reserves. The difference is explained by the signal type: AMM reserves accumulate gradually as liquidity providers withdraw, making them suitable for changepoint detection. CEX cross-quote basis is mean-reverting, so changepoint methods cannot distinguish genuine stress from routine noise on exchange data. Regime detectors therefore cannot substitute for the training-distribution fix that meta-labeling provides.

To test whether training diversity improves oracle capture, we run four conditions with a fixed calibration split (Terra/LUNA primary-signal windows) and a fixed test split (SVB). Table 15 summarises the results.

Table 15: Cross-mechanism training diversity (`basis_usdc_1m_gt10bps`, `price_plus_book`, threshold calibrated on Terra/LUNA, test on SVB).

Training set	Events	Net bps	Trades	Oracle cap.
Oracle (ceiling)	–	+162.2	315	100.0%
Terra/LUNA only	1	+78.5	123	48.4%
Celsius/3AC only	1	+79.7	221	49.1%
FTX only	1	+36.5†	1†	22.5%†
All four pooled	4	+83.7	163	51.6%

†Near-degenerate: FTX's credit shock produced < 5 bps USDC/USDT basis at Binance benchmark venues; the Terra-calibrated threshold (θ=0.61) filters all but one SVB window.

The pooled four-event training set spans 4,368 primary-signal windows across three mechanism classes (algorithmic, exchange-credit, regulatory), with positive rates ranging from 15.0% (BUSD) to 20.0% (Celsius/3AC). Oracle capture is stable at approximately 49% for single-event training on Terra/LUNA and Celsius/3AC, and rises to 51.6% with four-event pooling. This stability establishes that execution microstructure transfer is robust: depth withdrawal is a consistent precursor whose feature signature transfers across algorithmic, exchange-credit, and fiat-reserve mechanisms without requiring mechanism-matched training data.

FTX is the informative outlier. FTX's exchange credit shock was exchange-specific: USDC and USDT basis at Binance benchmark venues peaked at approximately −5 bps, far below the > 100 bps excursions in Terra/LUNA and Celsius/3AC. A model trained on FTX primary fires does not observe the depth-withdrawal feature signature in its training data, and the Terra/LUNA-calibrated threshold (θ=0.61) cannot recover a useful operating point. This is a mechanism-*intensity* rather than mechanism-*class* failure: exchange-credit shocks that do not propagate to benchmark-venue basis do not provide transferable training signal, regardless of mechanism label. The pooled model recovers from the FTX deficiency because Terra/LUNA and Celsius/3AC dominate the training distribution and provide sufficient depth-withdrawal signal.

The feature signature that drives transfer is consistent across conditions: ask-side depth and bid-ask spread are the leading microstructure predictors in the Terra/LUNA, Celsius/3AC, and pooled models, confirming that CEX market makers withdraw depth under uncertainty regardless of whether the underlying trigger is algorithmic collapse, exchange credit failure, or reserve-bank insolvency.

6.3.2 Density Is Necessary and Supervision Is Decisive. Meta-labeling and a conditioned PPO-GRU share identical training density (17% positive rate on primary-signal windows) yet differ by 111.7 bps in net return. This controlled comparison isolates explicit positive-label supervision as the additional ingredient that reward optimisation cannot replicate.

Table 16: RL policy versus supervised baselines (`basis_usdc_1m_gt10bps`, SVB test split).

Model (training)	Net bps	Trades
Oracle (hindsight ceiling)	+162.2	315
Meta-labeling (Terra/LUNA*)	+82.5	397
PPO-GRU conditioned† (Terra/LUNA*)	−29.2	919
PPO-GRU unconditioned (Terra/LUNA*)	degenerate	0
GRU supervised (calm-ctrl)	−239.0	656
MLP-Window supervised (calm-ctrl)	−340.7	1,052

*Cross-mechanism: Terra/LUNA train, SVB eval. †Conditioned on $|b_{\mathrm{USDC}}|$>10 bps windows, 17% positive rate.

The unconditioned PPO-GRU degenerates to zero entries, as the 2.3% positive rate in the full Terra/LUNA stream is too sparse to generate a reward signal that calibrates entry timing. Conditioning on primary-signal windows resolves the density problem: the agent trades (919 entries, AUROC 0.87) but earns −29.2 bps. The distinction between the conditioned agent and meta-labeling is qualitative. The meta-labeling classifier receives direct binary feedback, where y=1 marks exactly the profitable windows, while the RL agent must infer the same subset through a reward signal that is negative on 83% of entries. The supervision is categorically cleaner, not merely denser. The 919-trade overtrading pattern is itself informative. Policies that fail due to insufficient training converge toward NoTrade, not sustained false-positive entry. The 919-trade result is therefore more consistent with a timing-transfer mismatch from Terra/LUNA to SVB, meaning the agent learned Terra/LUNA's specific depth-withdrawal sequence and fires on it in the SVB context even when the timing does not match. This interpretation has a constructive implication. If the failure reflects a mechanism mismatch rather than an RL capability limit, then multi-mechanism training that exposes the agent to depth withdrawal events across diverse stress triggers should improve selectivity without requiring a fundamentally different RL algorithm. Combining the cross-mechanism protocol established here for supervised transfer with a reactive simulator that provides market-impact feedback is the natural next step.

7 Limitations

The BTC-USDC buy leg uses kline-proxy depth for the SVB window, since the perpetual contract did not exist on Binance until 2024. A 20% depth haircut confirms robustness (Table 7). The same kline-proxy applies to all four training events (Terra/LUNA, Celsius/3AC, FTX, BUSD), as the BTC-USDC perpetual did not exist during any of these windows; the 20% haircut sensitivity applies equally across all training conditions.

We validate the δ=5 bps settlement penalty assumption using on-chain Etherscan data for 100,000 USDC transfers across the SVB window (Table 17). The median ERC-20 transfer cost on 10K notional was 1.4–3.2 bps on nine of ten days, confirming that δ=5 bps is conservative. The exception is March 11 (peak depeg), where gas spiked to 61.5 gwei and the P95 cost reached 8.6 bps, the only day the assumption is tight. Burn events (institutional USDC redemptions to Circle) peaked on March 13 (29 events) and transfer volume hit $2.4B on that day, providing on-chain corroboration of the depth depletion visible in the CEX order book during the same window.

Table 17: On-chain settlement cost validation (USDC ERC-20 transfers, 100K sample, Mar 10–20 2023, ETH ≈ $1,580). δ=5 bps is conservative on all days except the Mar 11 gas spike at peak depeg.

Date	Med gas (gwei)	Med cost (bps)	P95 cost (bps)	Burns
Mar 10	30.8	3.16	5.70	4
Mar 11	61.5	6.31	8.64†	16
Mar 12	21.6	2.21	3.68	22
Mar 13	26.1	2.68	4.47	29
Mar 14–19	13.9–24	1.43–2.46	2.44–4.11	0–2

†Only day where P95 cost exceeds δ=5 bps. Gas data from 100K USDC transfers via Etherscan API.

Tier-A coverage is currently limited to the SVB event; extending to AMM or exchange-credit events requires route-complete L2 archives that are not yet available.

The RL results carry two important caveats. First, the conditioned PPO-GRU's −29.2 bps result may reflect overfitting to Terra/LUNA's specific depth-withdrawal timing rather than a fundamental RL limitation, which additional cross-mechanism training pairs could resolve. Second, unlike reactive-simulator approaches such as Olby et al. [23], the PPO-GRU is trained on historical replay without market-impact feedback. In a thin-book stress regime this systematically miscalibrates entry sizing because the agent observes book depth as it was rather than as it would be after its own orders clear liquidity, which is a plausible contributor to the 919-trade overtrading pattern. Integrating the cross-mechanism protocol shown here into a reactive simulator is the most direct path to resolving the timing component of the oracle gap.

8 Conclusion

The 12× optical-to-executable gap establishes that stablecoin arbitrage benchmarks built on price labels measure the wrong quantity. The executable fraction of apparent opportunities is small, shrinks with notional size, and is invisible to every model trained on calm-market data. Cross-mechanism training on a structurally different stress event is the only empirically validated path to positive net returns, and the RL diagnostic provides a precise account of why: label density enables trading, but binary supervision over the profitable subset provides the directional accuracy that reward optimisation cannot replicate at low positive rates. The benchmark, labels, oracle implementation, and training scripts are released under CC-BY 4.0 (URL redacted for review). StressBench is designed to support five lines of future work: execution-aware dislocation detection under cost uncertainty; cross-mechanism stress transfer to new event types as Tier-A coverage expands; RL execution policy learning with the cross-mechanism initialisation protocol established here; reactive simulation of thin-book stablecoin markets, where the oracle ceiling provides a principled performance target; and multi-mechanism RL training using the pooled-event protocol established here, which is the direct path to closing the 111.7 bps gap between supervised meta-labeling and the conditioned RL agent. The benchmark harness additionally ships four practitioner desk-rule baselines (route-consistent, depth-adjusted, spread-depth, and latency-haircut rules) that exercise the same evaluation pipeline and serve as interpretable reference points for future model comparisons.

Acknowledgements

Anonymised for review.

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